Protein Domain Structure Classification

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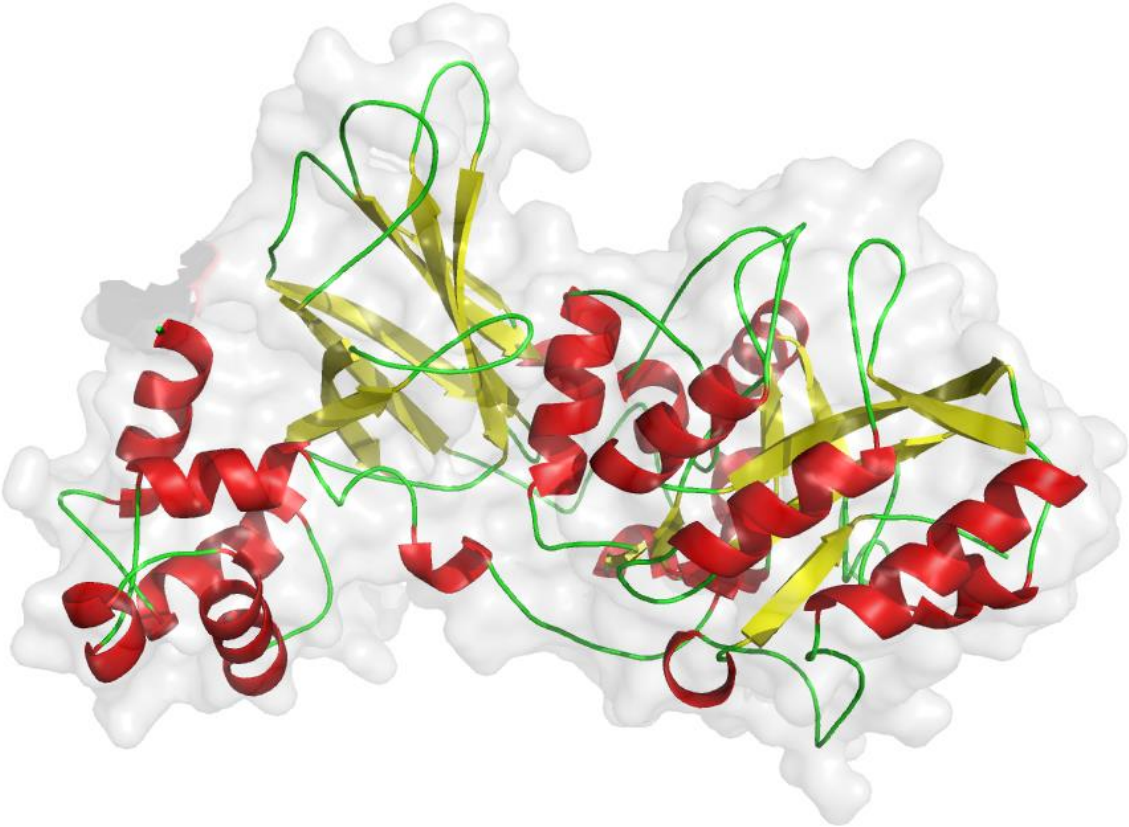
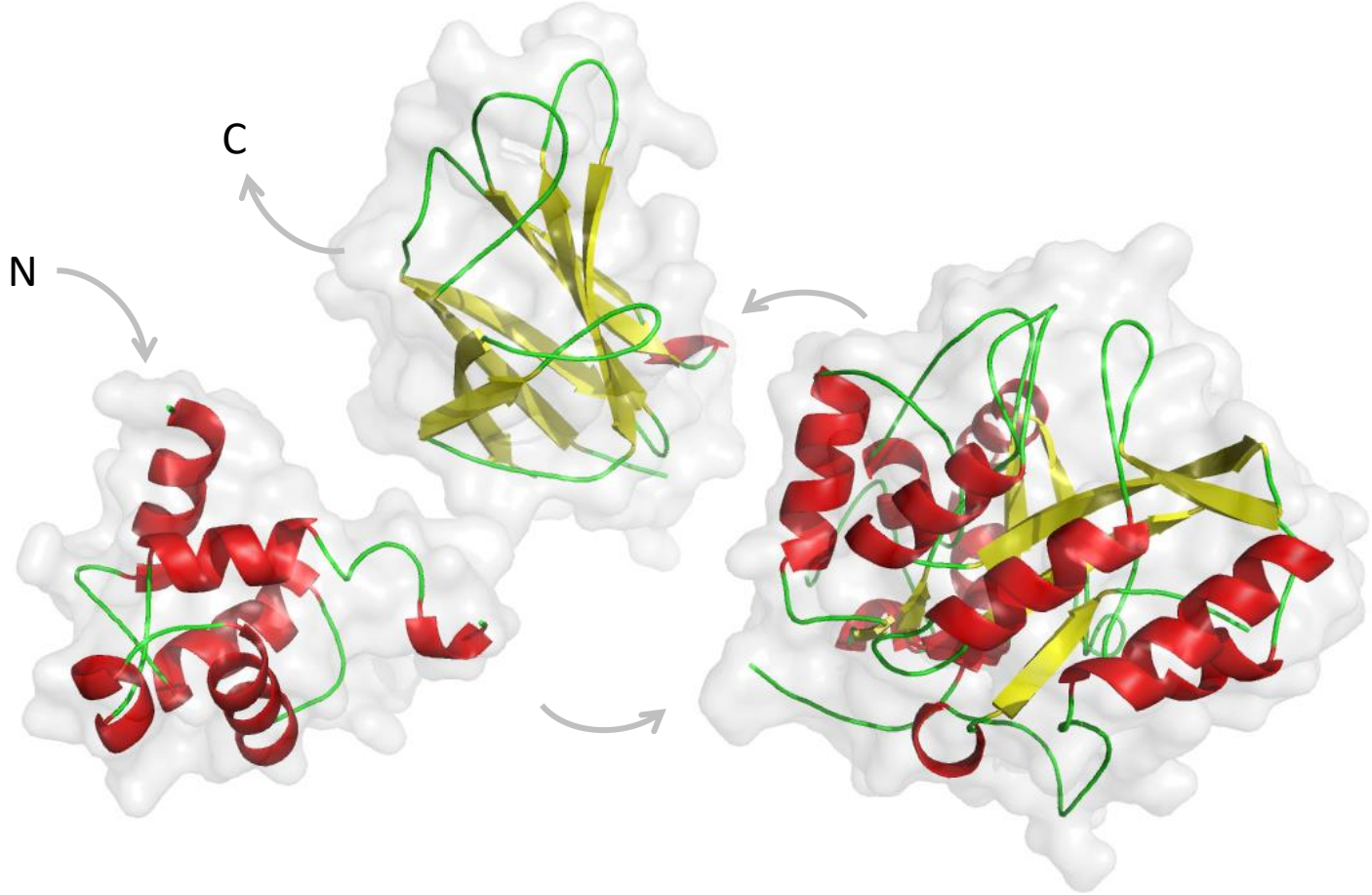
What is a protein domain?

- compact, semi-independent folding unit of protein structure

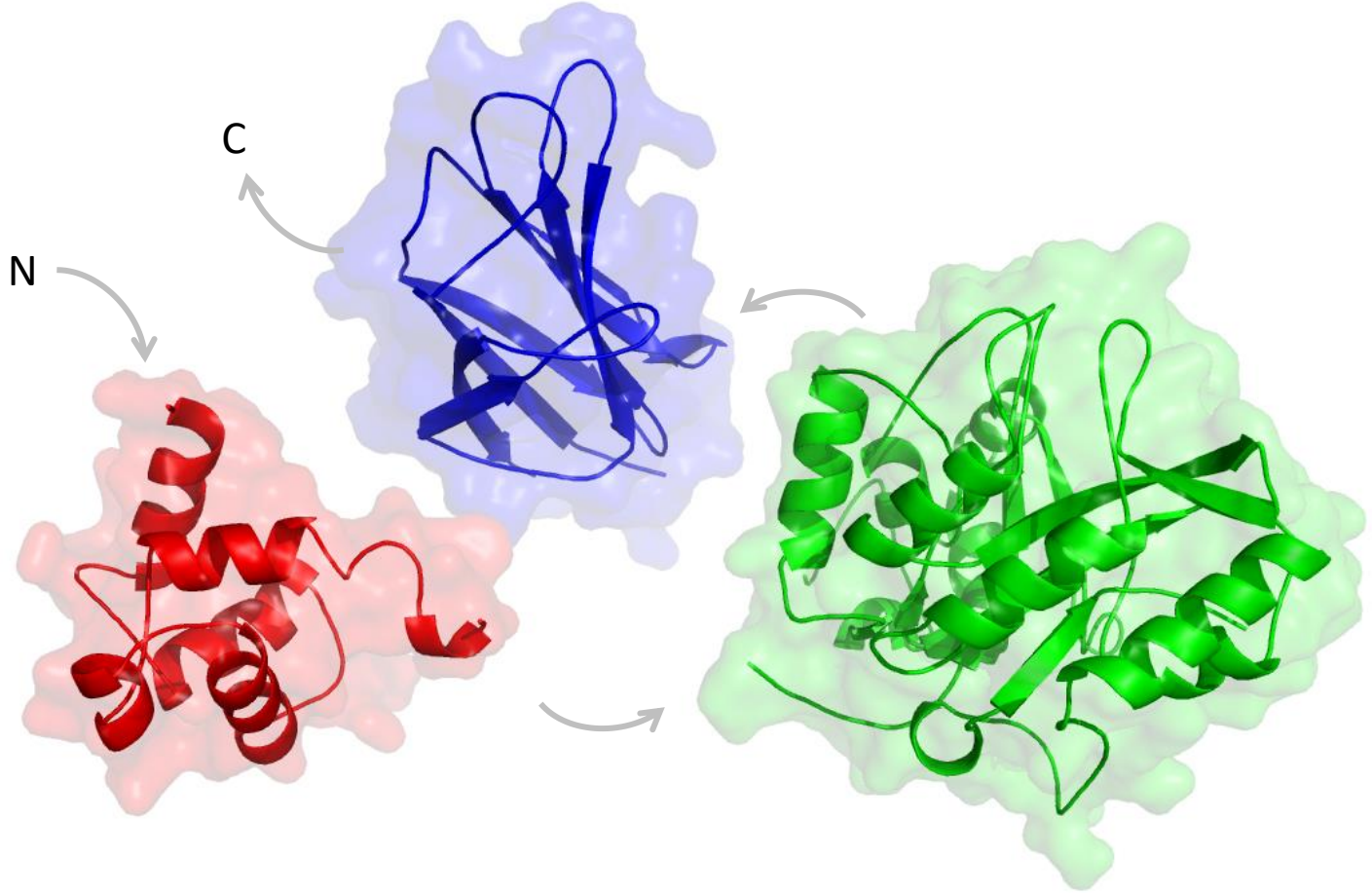
Structural Domains



Structural Domains

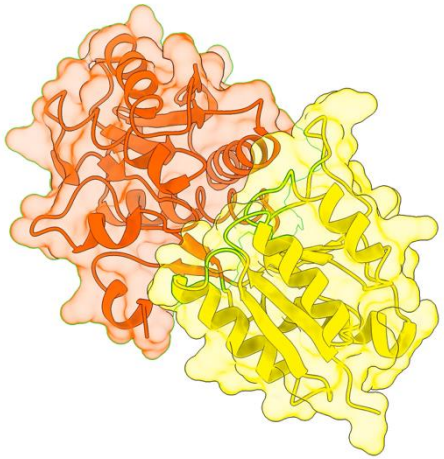


Structural Domains



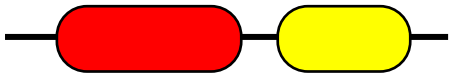
Domains are important evolutionary units

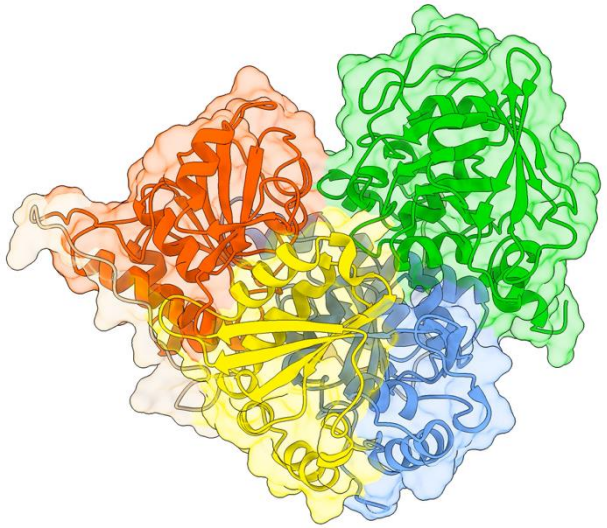
- 60-80% of known genes code for multidomain proteins



Arabinose Binding protein

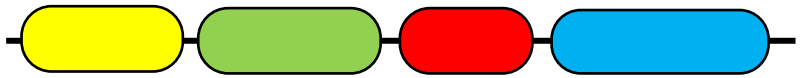
2 domains





Aconitase

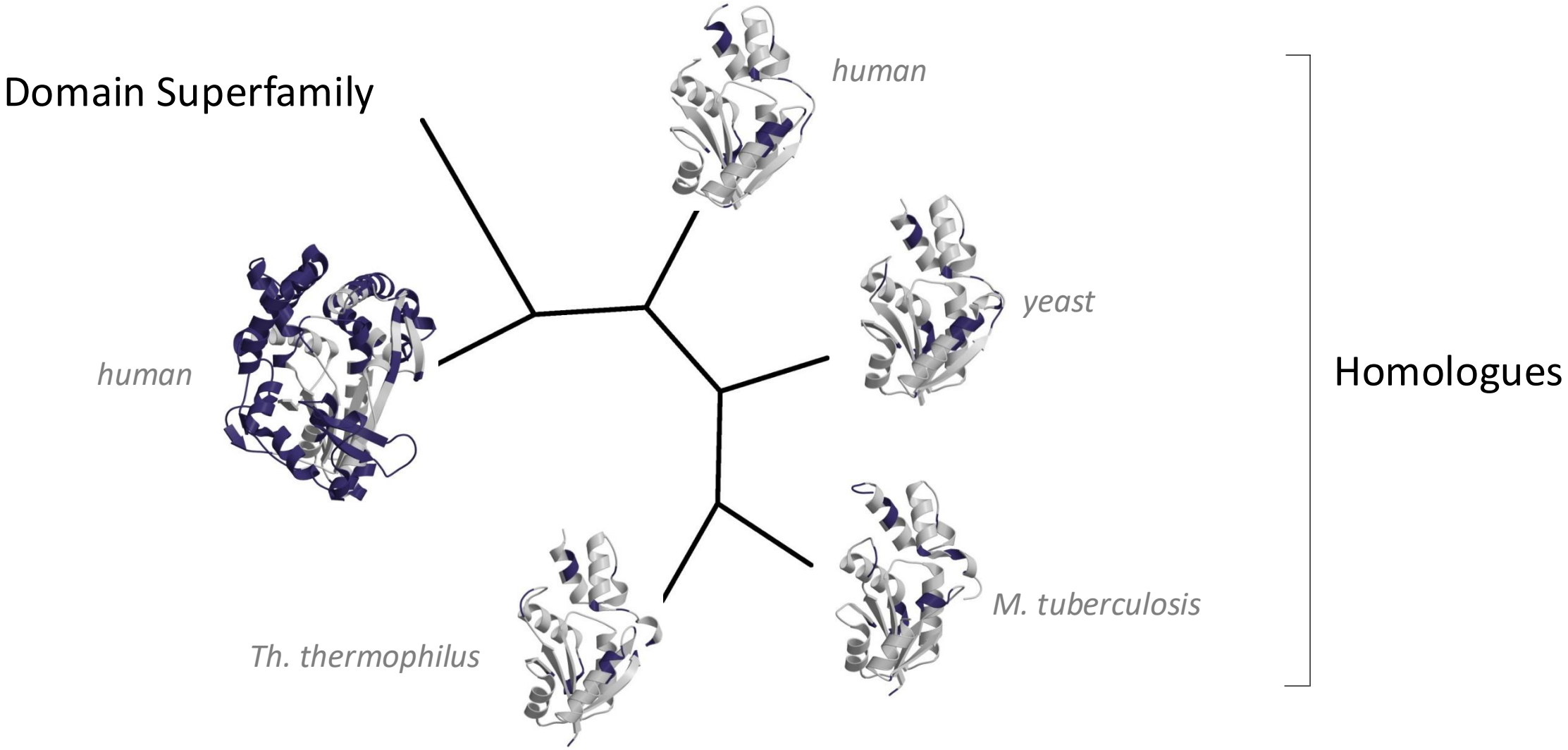
4 domains



What is a protein domain superfamily?

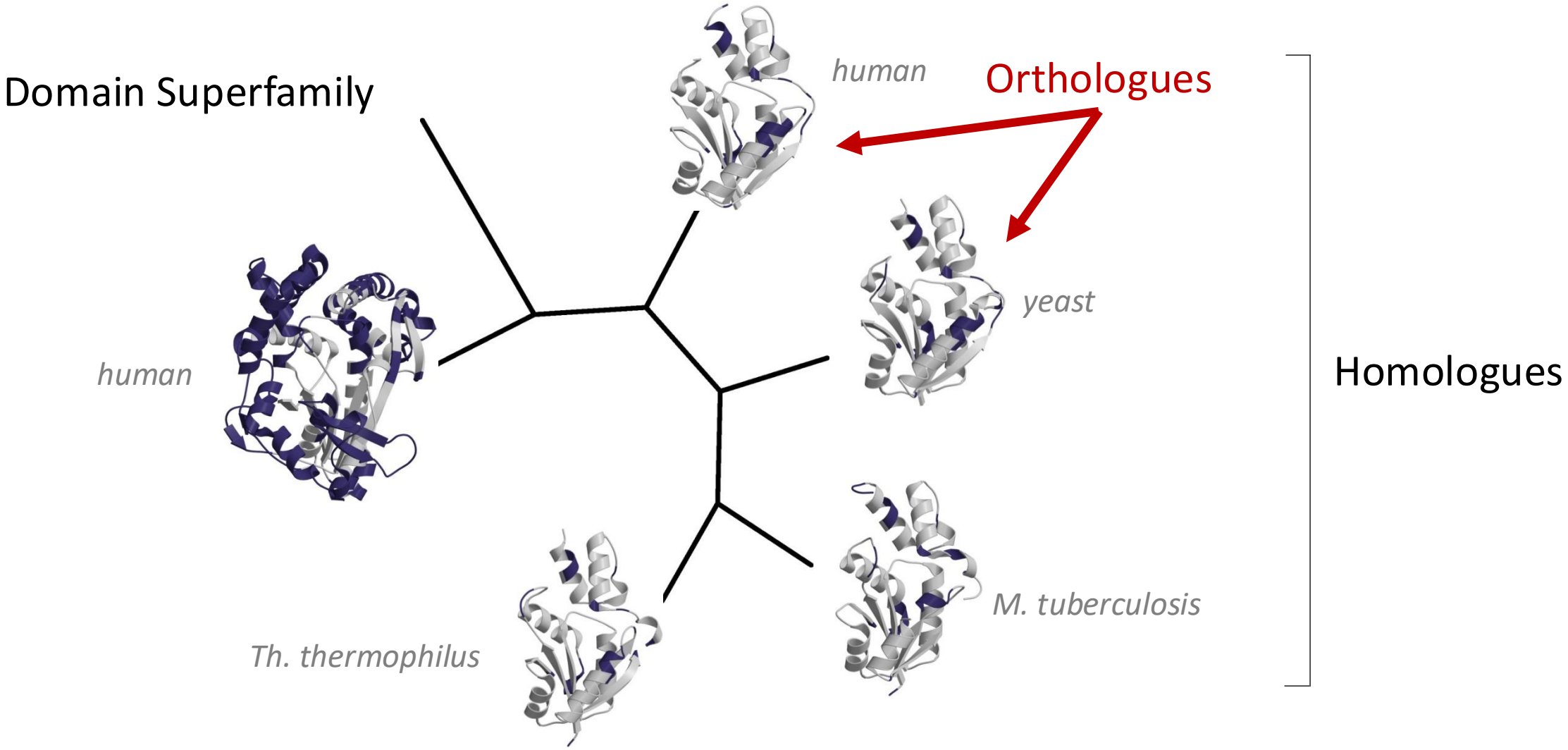
- group of protein domains that share a common evolutionary ancestor

Evolution gives rise to families of proteins



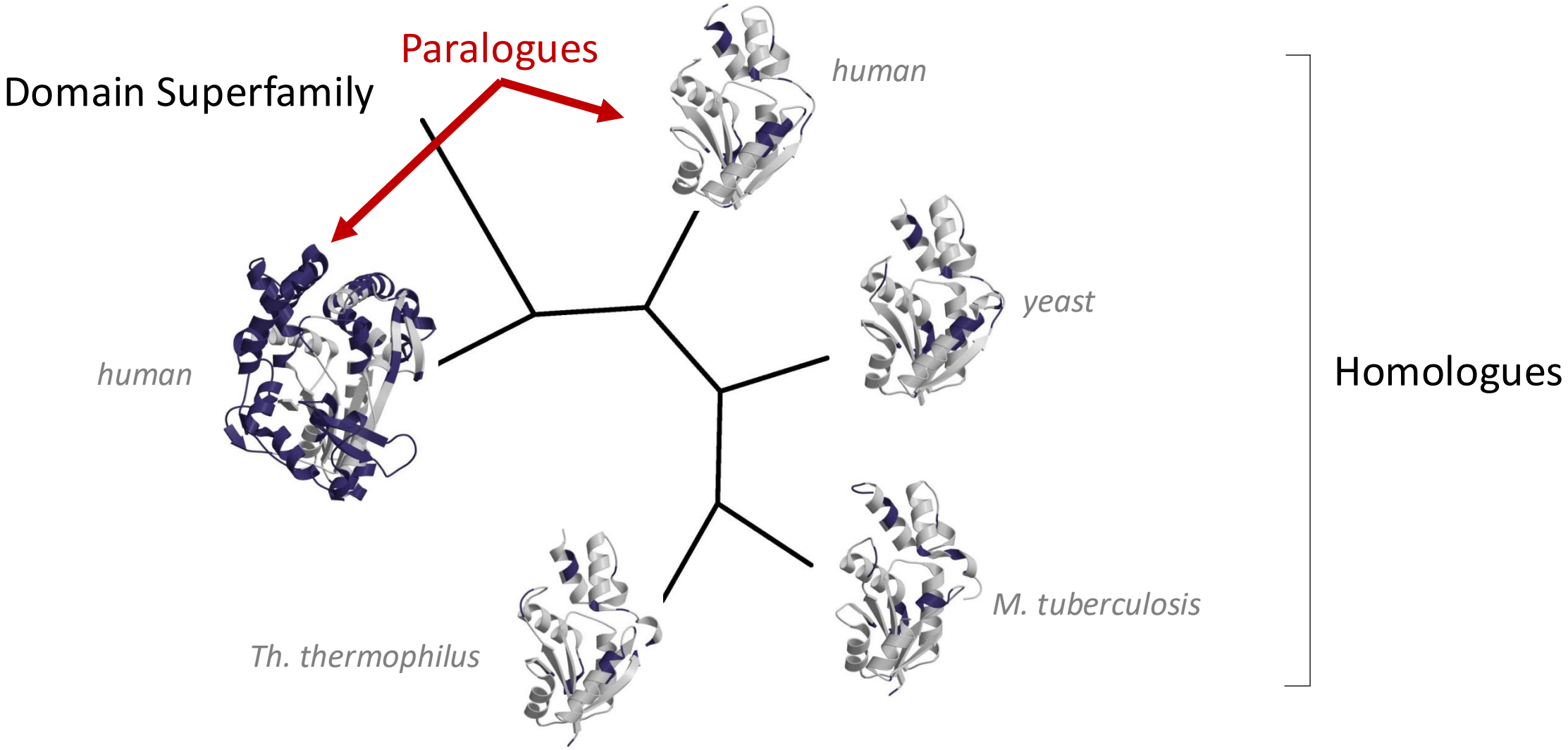
Structure is more conserved than sequence during evolution

Evolution gives rise to families of proteins



Structure is more conserved than sequence during evolution

Evolution gives rise to families of proteins

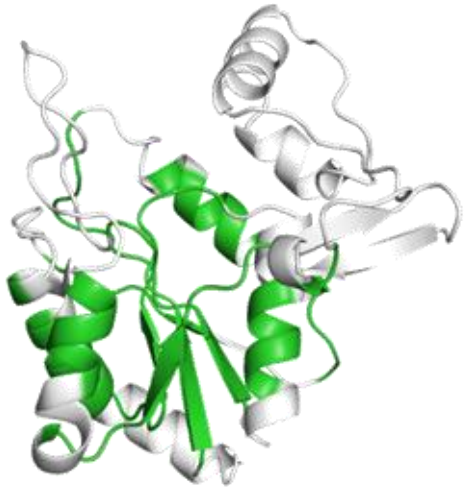


Structure is more conserved than sequence during evolution

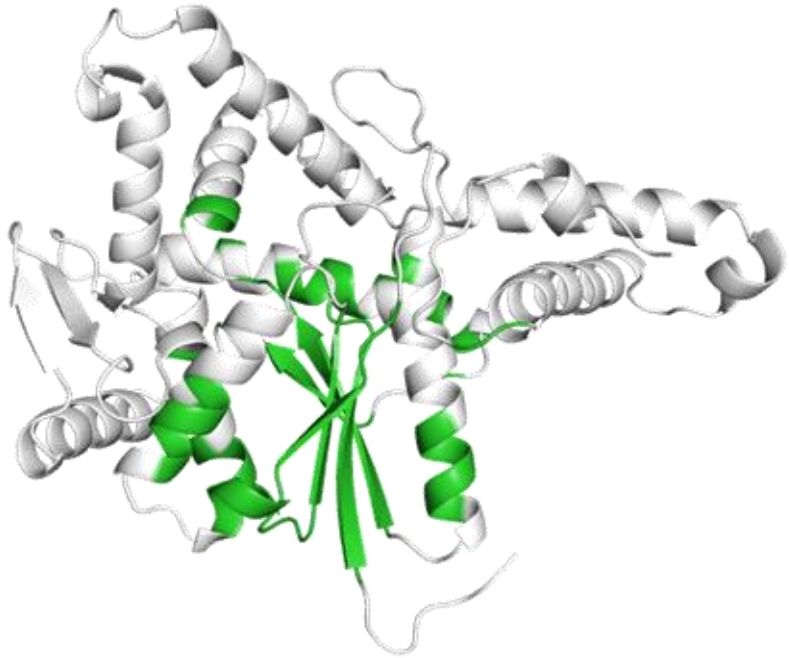
Structural diversity in CATH Superfamilies

- Structure is more highly conserved than sequence during evolution
- At least 40-50% of the structure is conserved

Universal stress protein A



Arginine tRNA ligase



Sulfate adenylyltransferase

Protein structure classifications

Grouping protein domains based on common evolutionary ancestors

CATH: Domain Classification Database

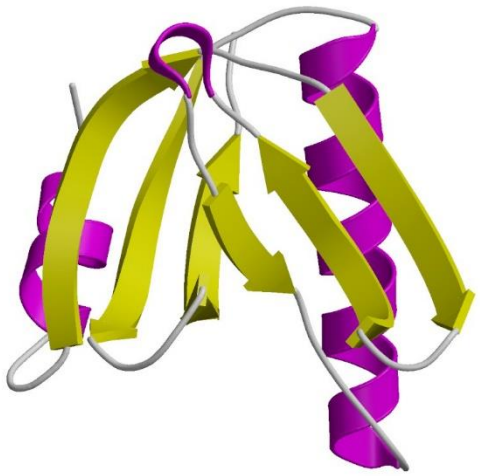
- Orengo & Thornton (1993)
- Classifies ~500,000 domains into ~5500 domain superfamilies

Class

Architecture

Topology (or fold group)

Homologous Superfamily



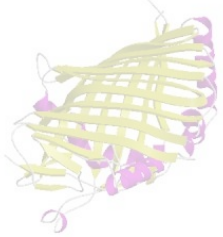
CATH Superfamily: 2.40.50.50

Hierarchical Structural Classification

Class

1: Mostly Alpha

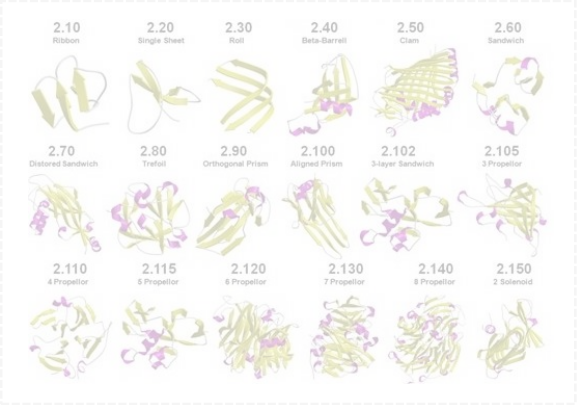
2: Mostly Beta

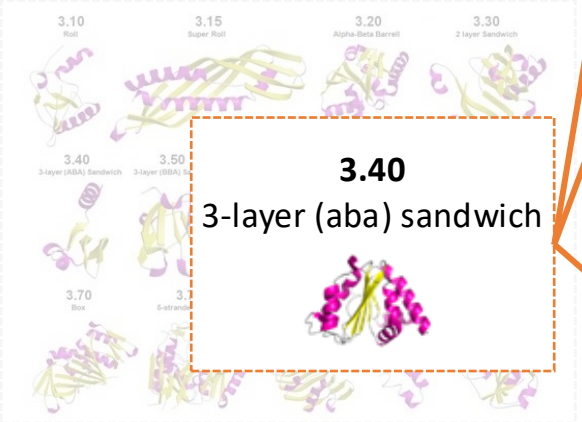


3: Mixed Alpha-Beta

4: Few Secondary Structures

Architecture

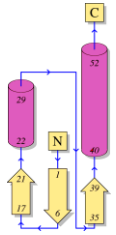




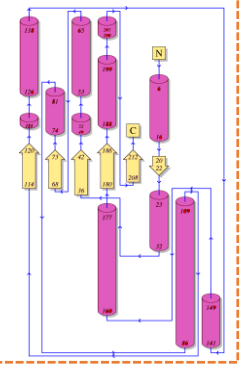
3.40
3-layer (aba) sandwich

Topology

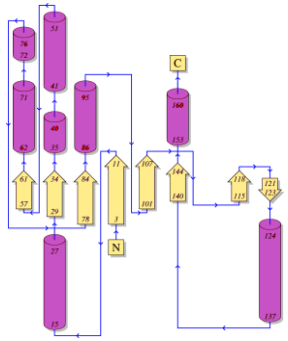
3.40.5
Ribosomal Protein L9



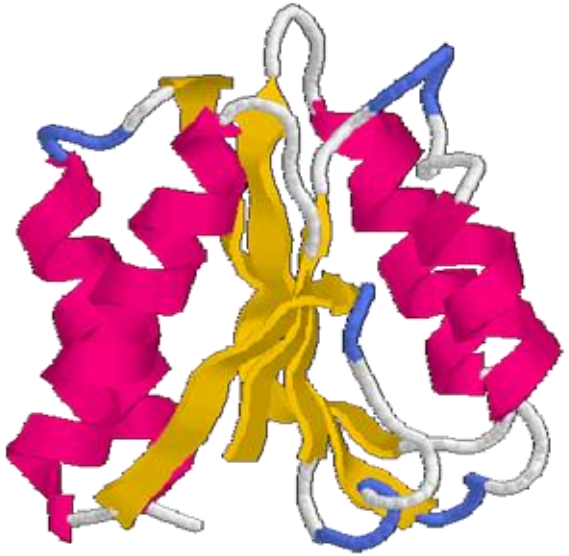
3.40.35
Fructose Permease



3.40.50
Rossmann

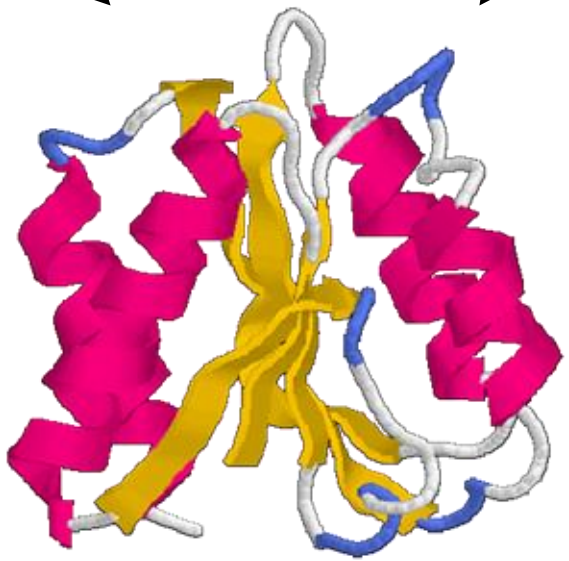


Divergent Evolution



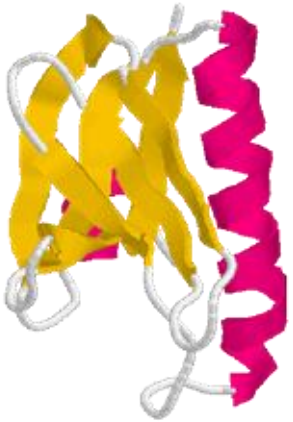
..VILST.. ..KLPST.. ..SLTRF..

..VILST.. ..KLPST.. ..SLTRF..



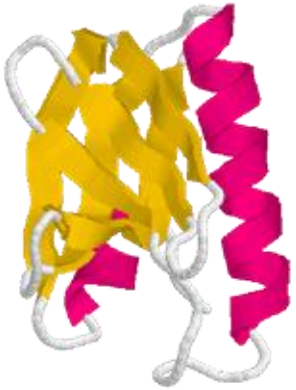
Convergent Evolution

Homologous Structures



cholera toxin

pertussis toxin

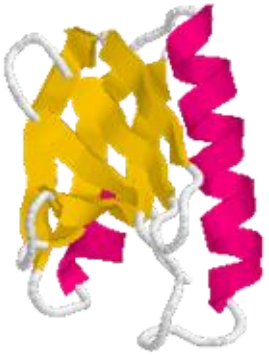


Heat labile enterotoxin

97	**81**	**SSAP score**
79%	**12%**	**Sequence identity**

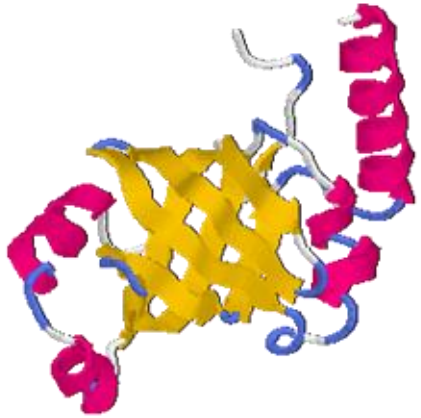
- high structure similarity score, often < 4A
- may have detectable sequence similarity e.g. by HMMs
- may have related functions

Analogous Structures



Heat labile enterotoxin

Staphylococcal nuclease

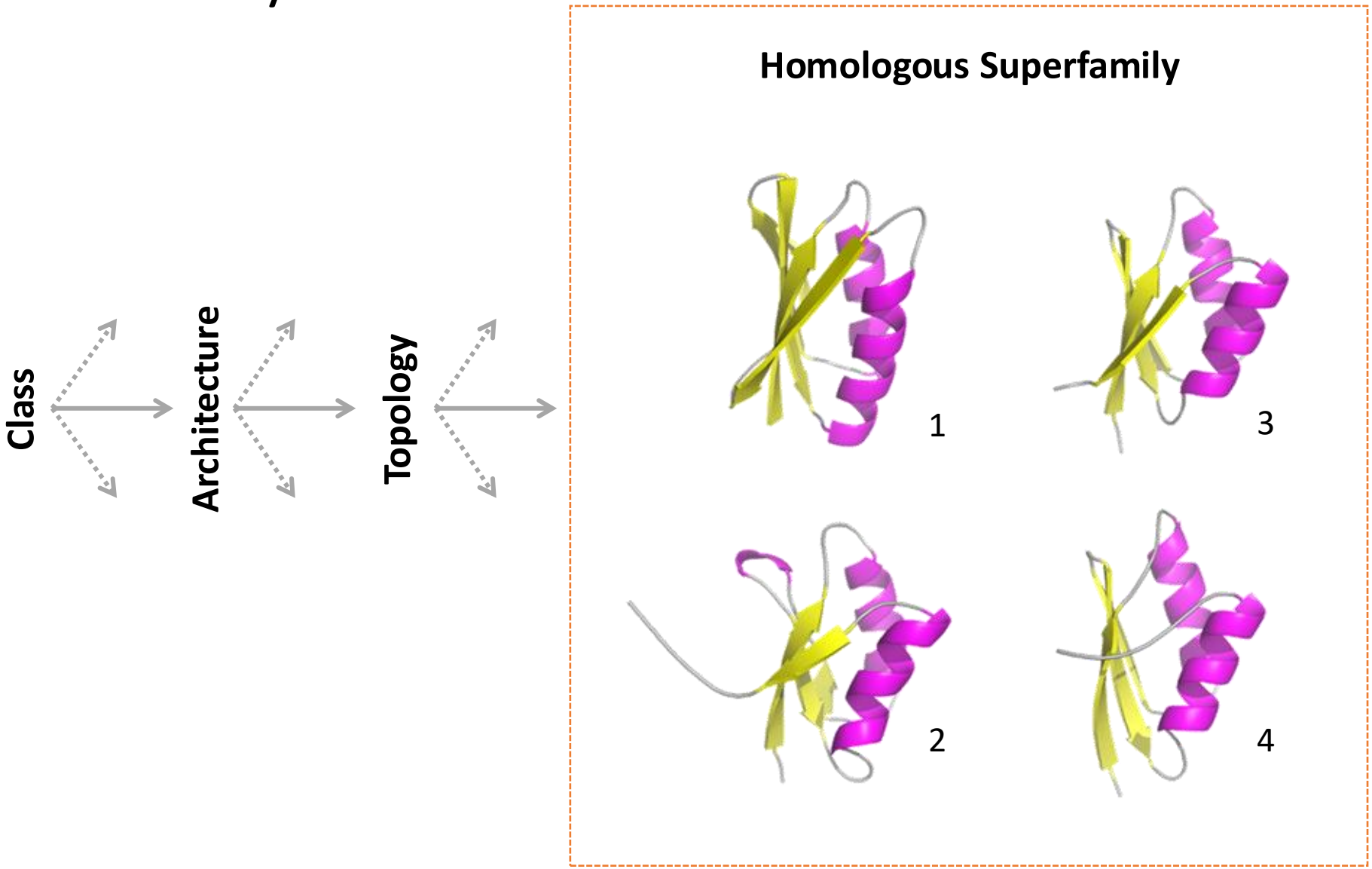


Lysyl TRNA synthetase

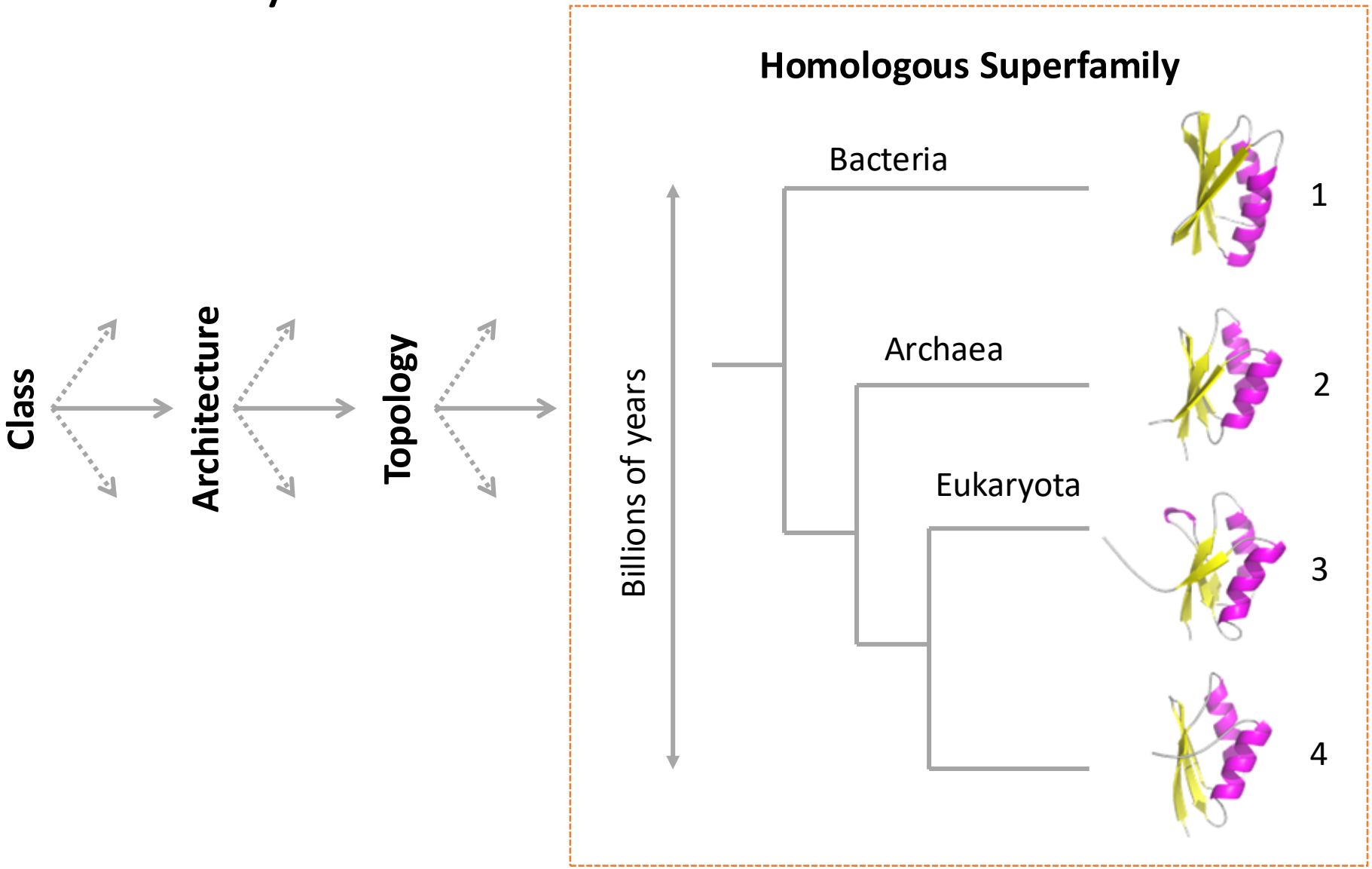
	Staphylococcal nuclease	*Lysyl TRNA synthetase*
SSAP score	**71**	**75**
Sequence identity	**8%**	**5%**

- structure similarity
- no sequence similarity
- no functional similarity

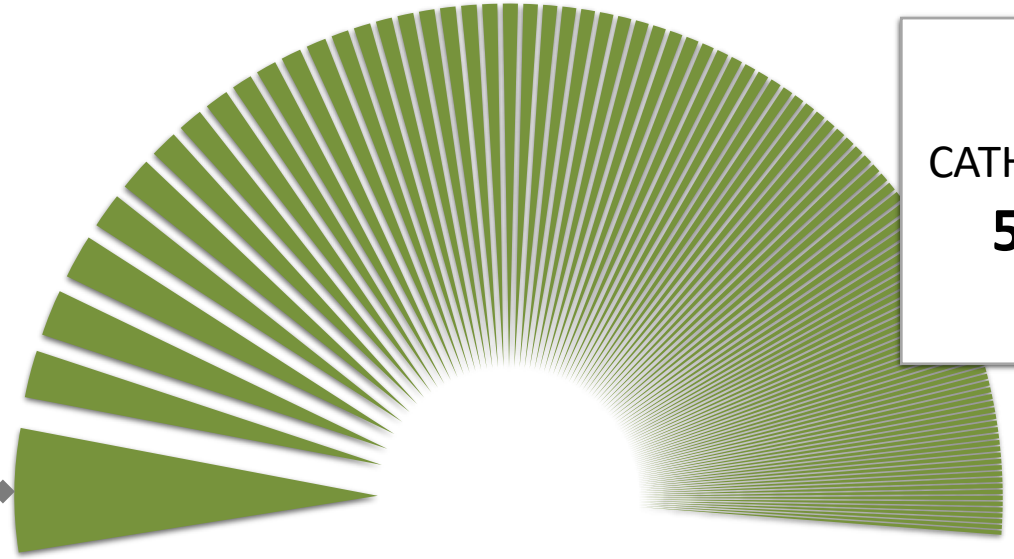
Superfamily-centric Classification



Superfamily-centric Classification

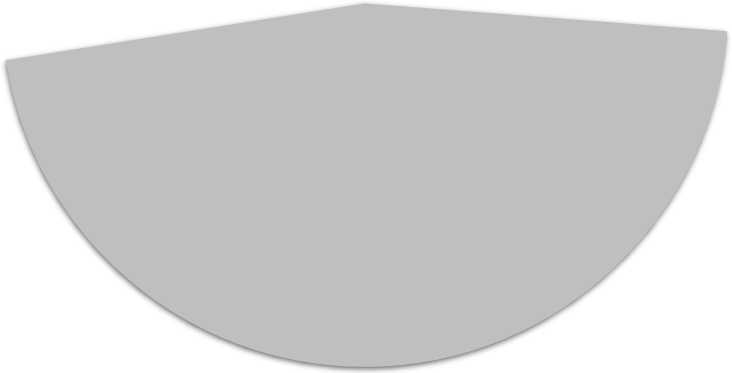


Distribution of CATH superfamilies



100 most populated CATH-Gene3D superfamilies contain **54%** of the **200+ million** known domain sequences

The largest CATH-Gene3D superfamilies can encompass a huge amount of sequence, structure and functional diversity



Remaining **~5000** CATH-Gene3D superfamilies contain **46%** of all known domain sequences

Protein Domain Family Classifications

SCOP

Experimental domain structures
Alexey Murzin, LMB, Cambridge

SUPERFAMILY

Predicted domain structures
Julian Gough, Bristol University

CATH-Gene3D

Experimental domain structures
Predicted domain structures
Christine Orengo, UCL

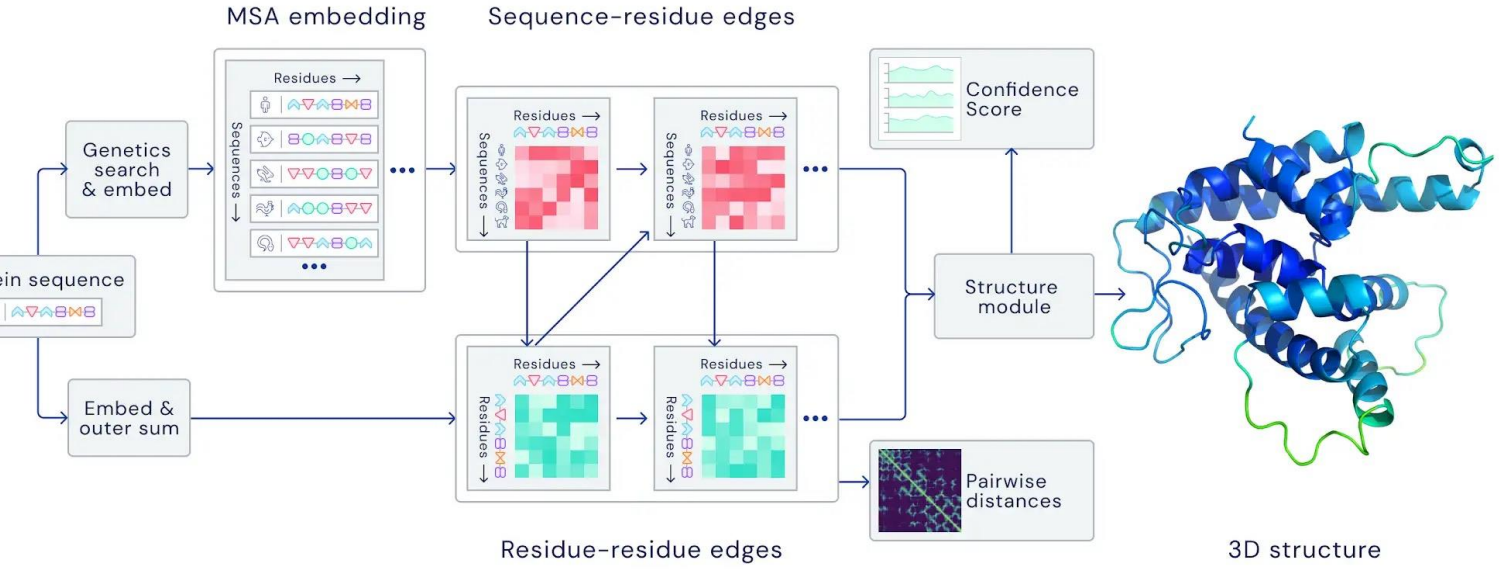
ECOD

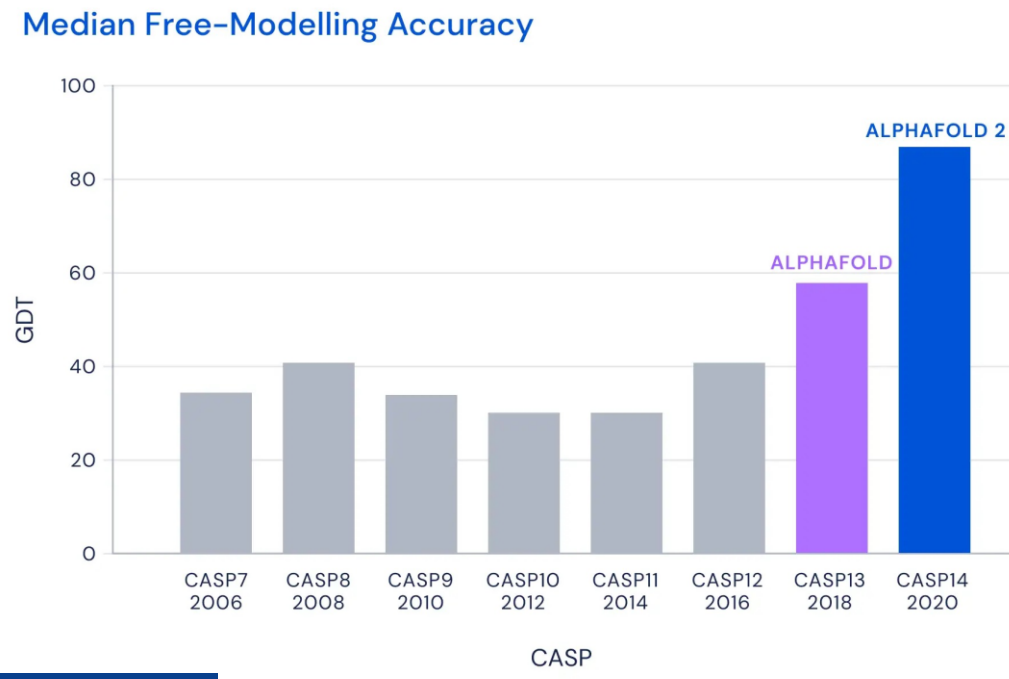
Experimental Domain Structures
Grishin, UTSW

AlphaFold + TED

Accurate protein structure prediction

AlphaFold2 and AlphaFold Database

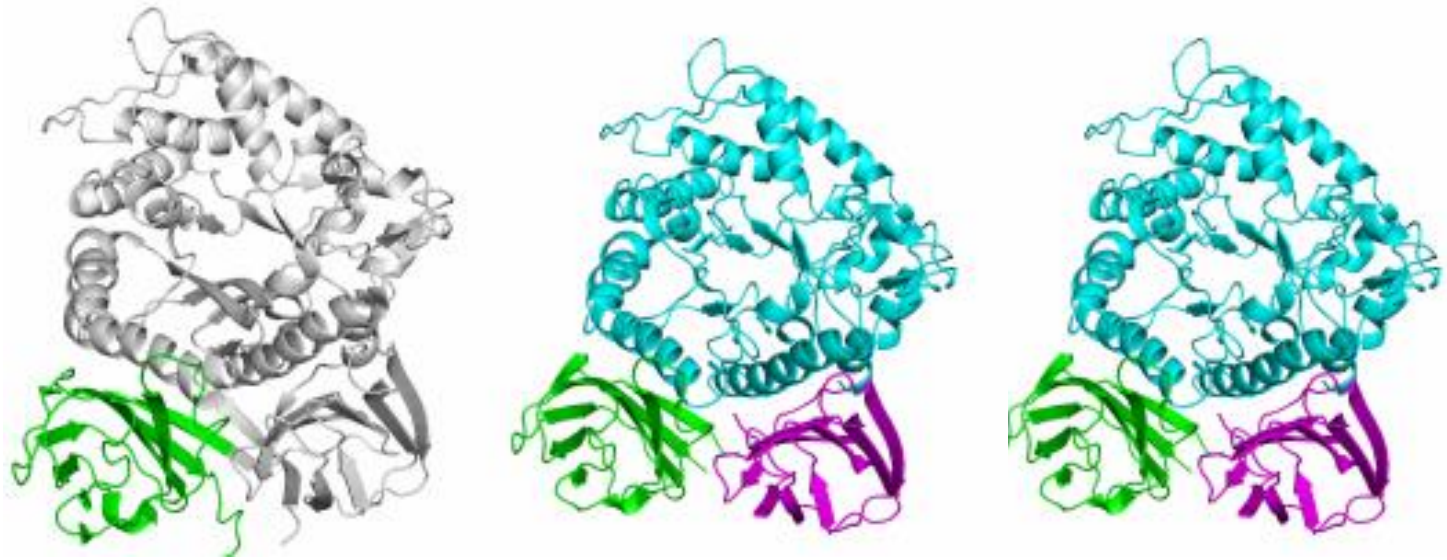




200 million protein structure models!

Consensus domain segmentation

188 million predicted **protein** structures



Unidoc

Chainsaw

Merizo

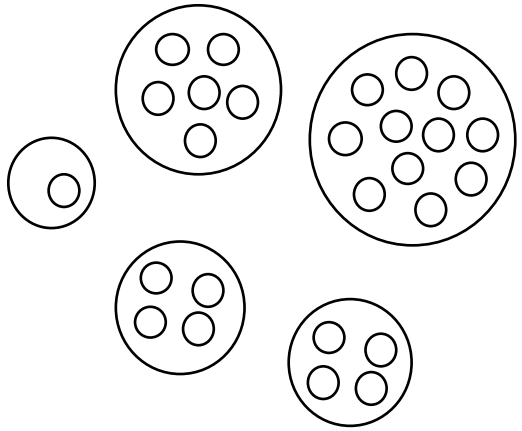
365 million predicted **domain** structures

(100 million more than found by sequence)

If **2 out of 3 methods agreed** we accepted the boundaries

CATH Assignment pipeline

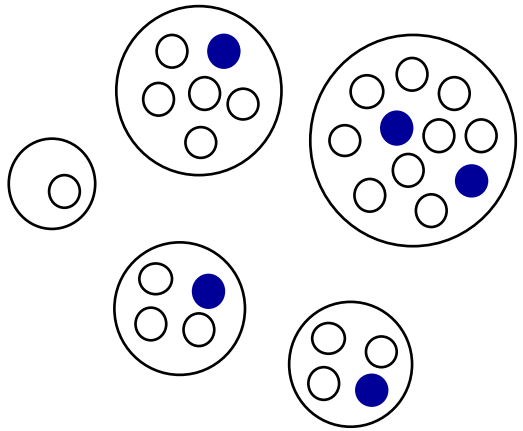
365 million domains



121 million clusters (50% seq. id.)

FoldSeek comparisons against CATH PDB domains

CATH match



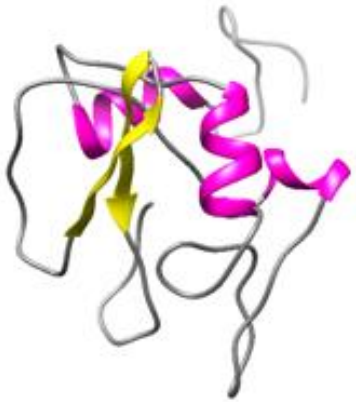
77% of domains (>251 million) assigned to CATH

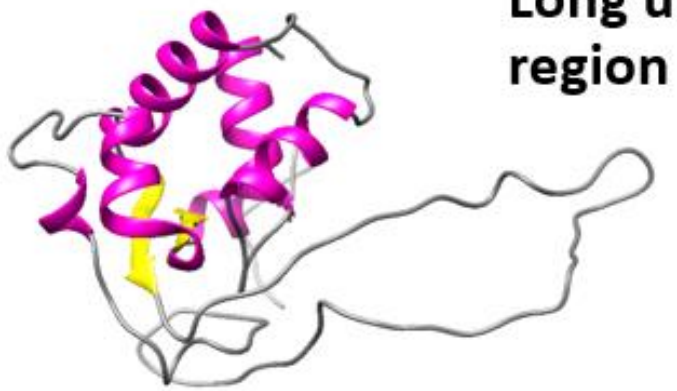
CATH expanded >400-fold

Lau, Bordin et al. Science (2024)

Removing models with problematic features for domain structure classification in CATH

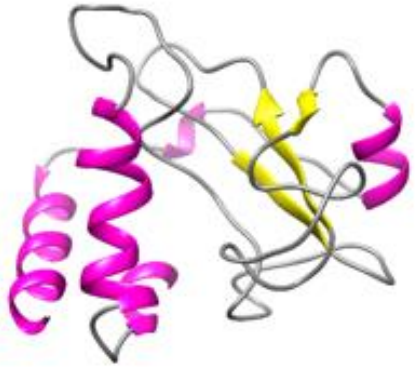
Highly unordered > 65% of domain

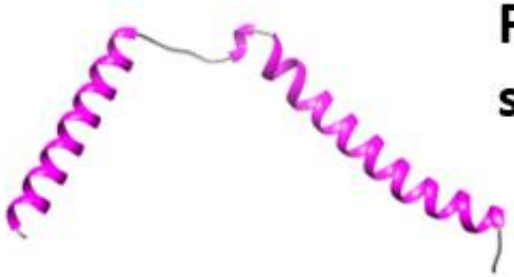




Long unordered region

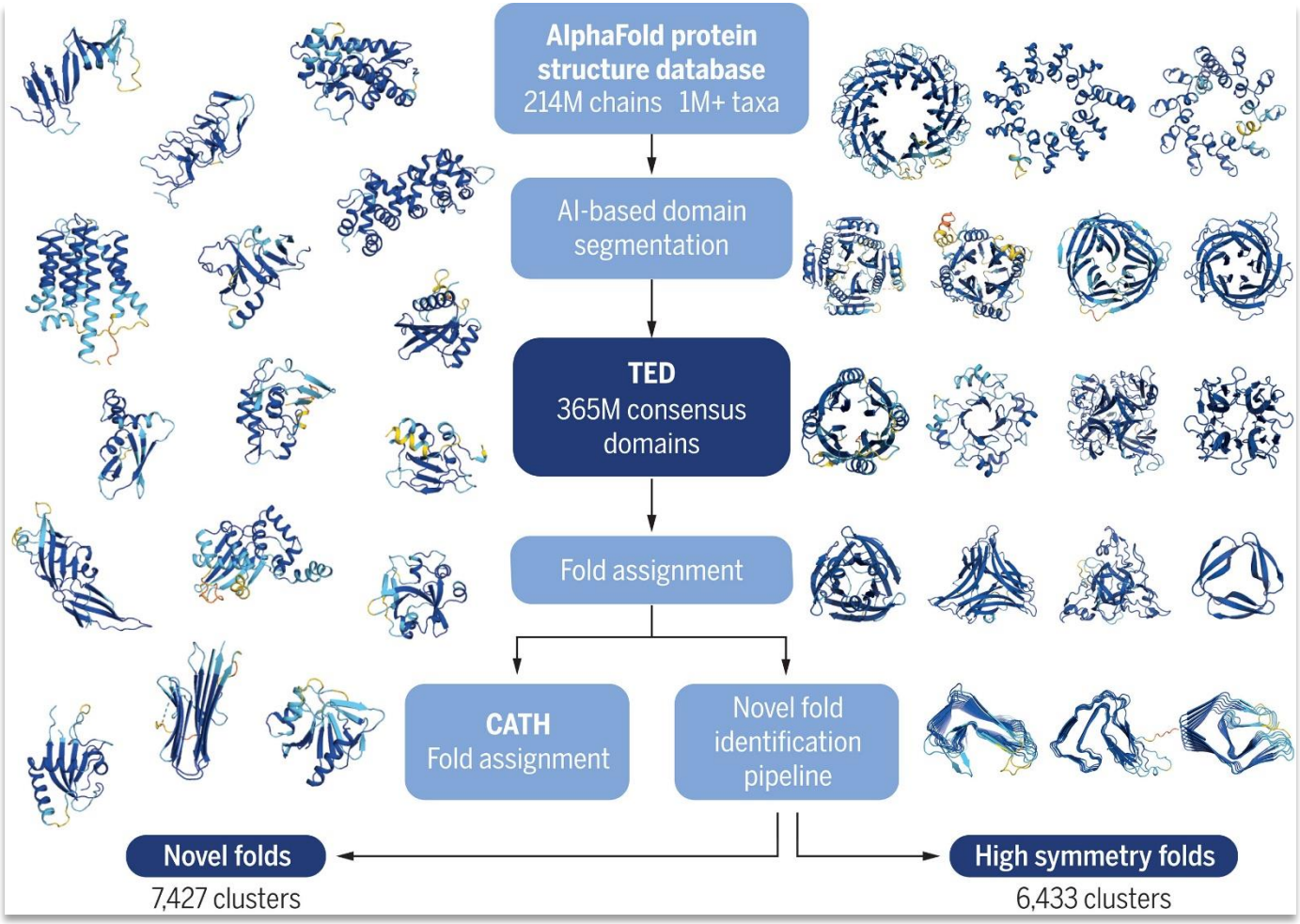
Poorly packed



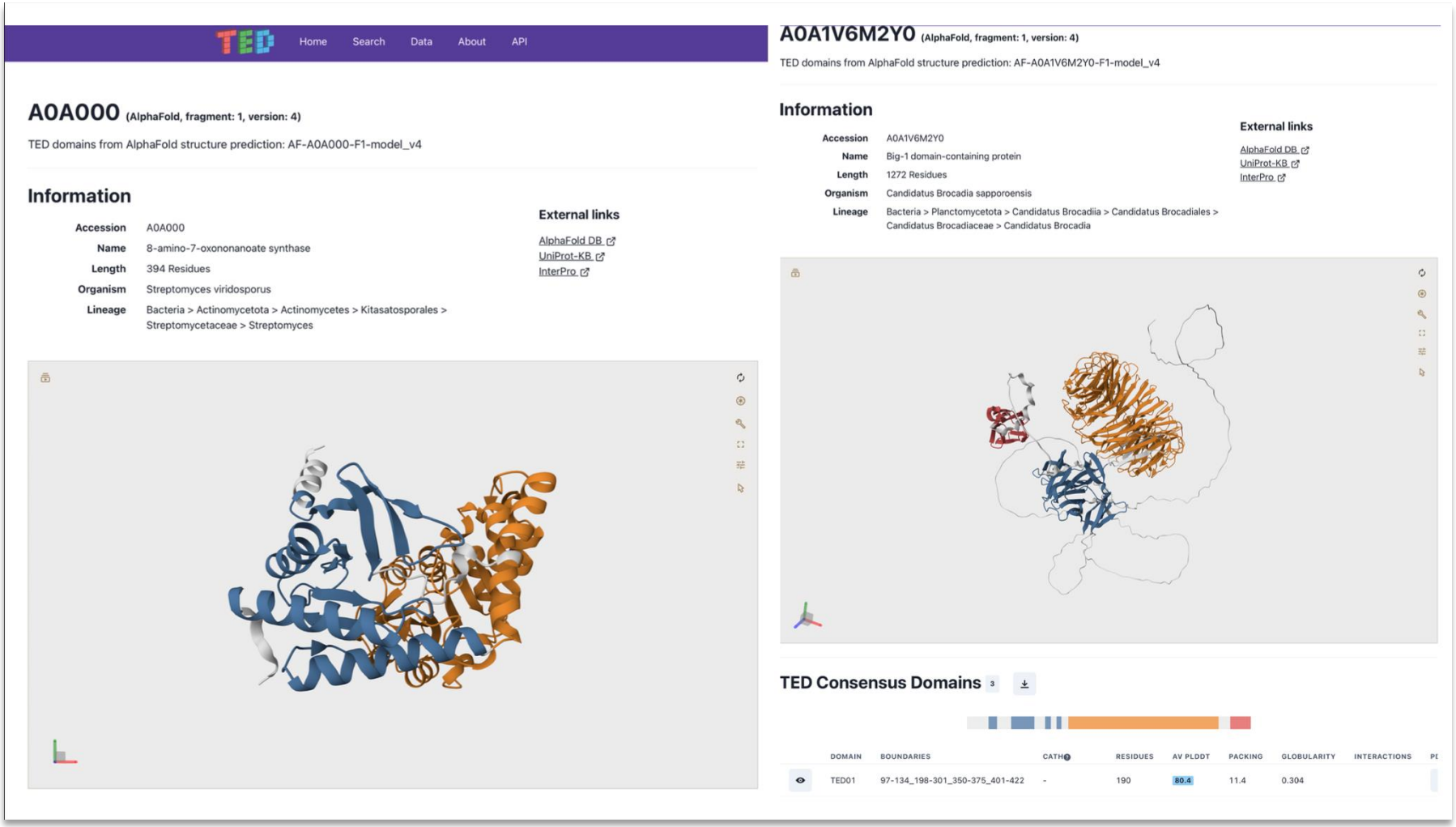


Few secondary structures

The Encyclopedia of Domains (TED)



Andy M. Lau *et al.*, Exploring structural diversity across the protein universe with The Encyclopedia of Domains. *Science* **386**, eadq4946(2024). DOI:10.1126/science.adq4946



https://ted.cathdb.info

Dissemination of TED domain data via AFDB

Collaboration with PDBe at EBI

Sameer Velankar

Quality score **TED Qscore**

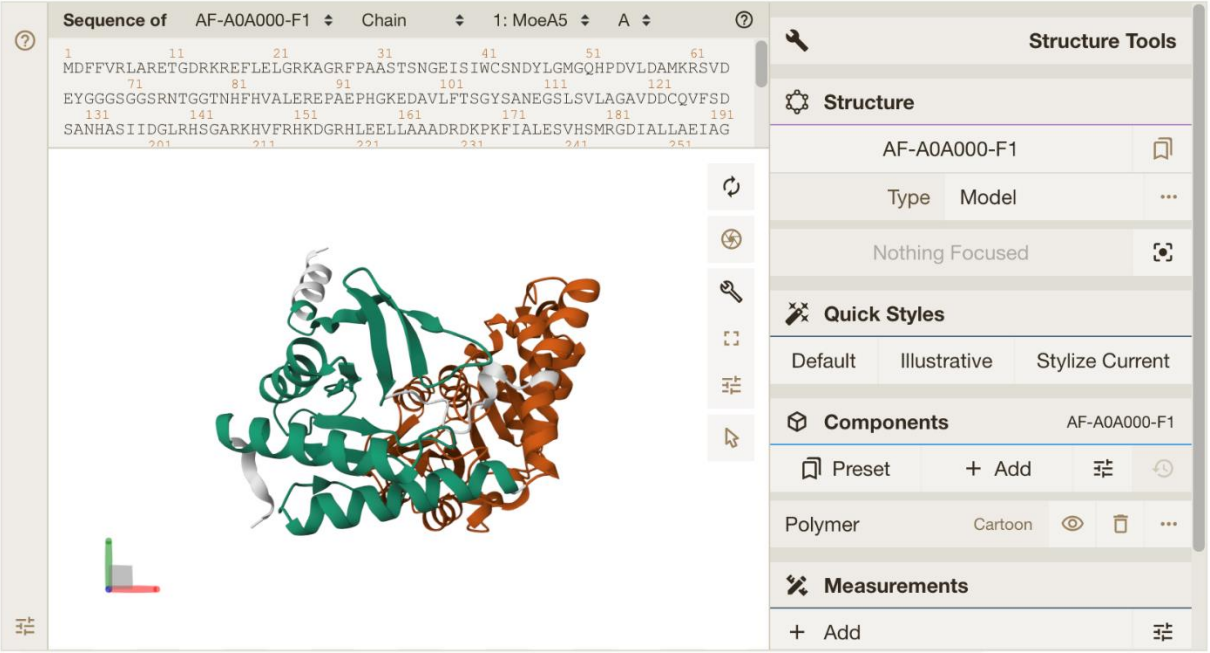
$$Q = 100 * (4 * \min(C1,C2) + TC + CO + PL + EV') / 8$$

EV’	structural similarity score to a CATH domain
C1, C2	bidirectional residue overlaps
CO	compactness
TC	agreement between domain segmentation methods
PL	pLDDT

Structure viewer

TED domain data in AFDB

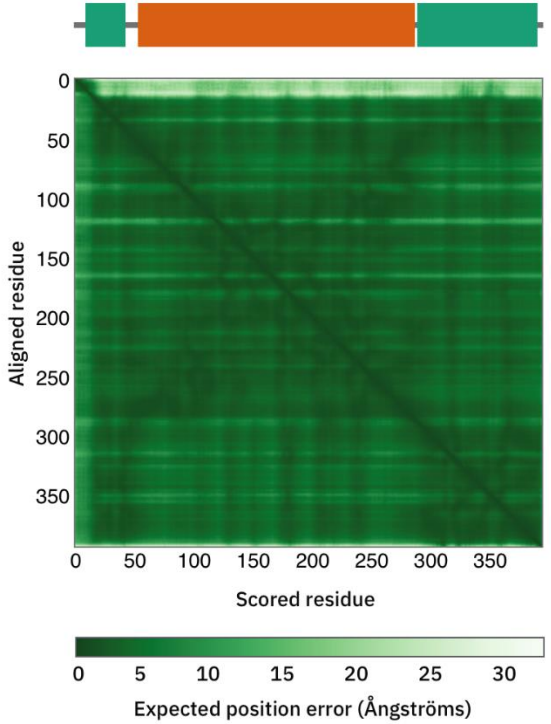
Mol* 3D Viewer



Model Confidence | TED Domains

Domain 1 Domain 2

TED Domains and Predicted Aligned Error (PAE)

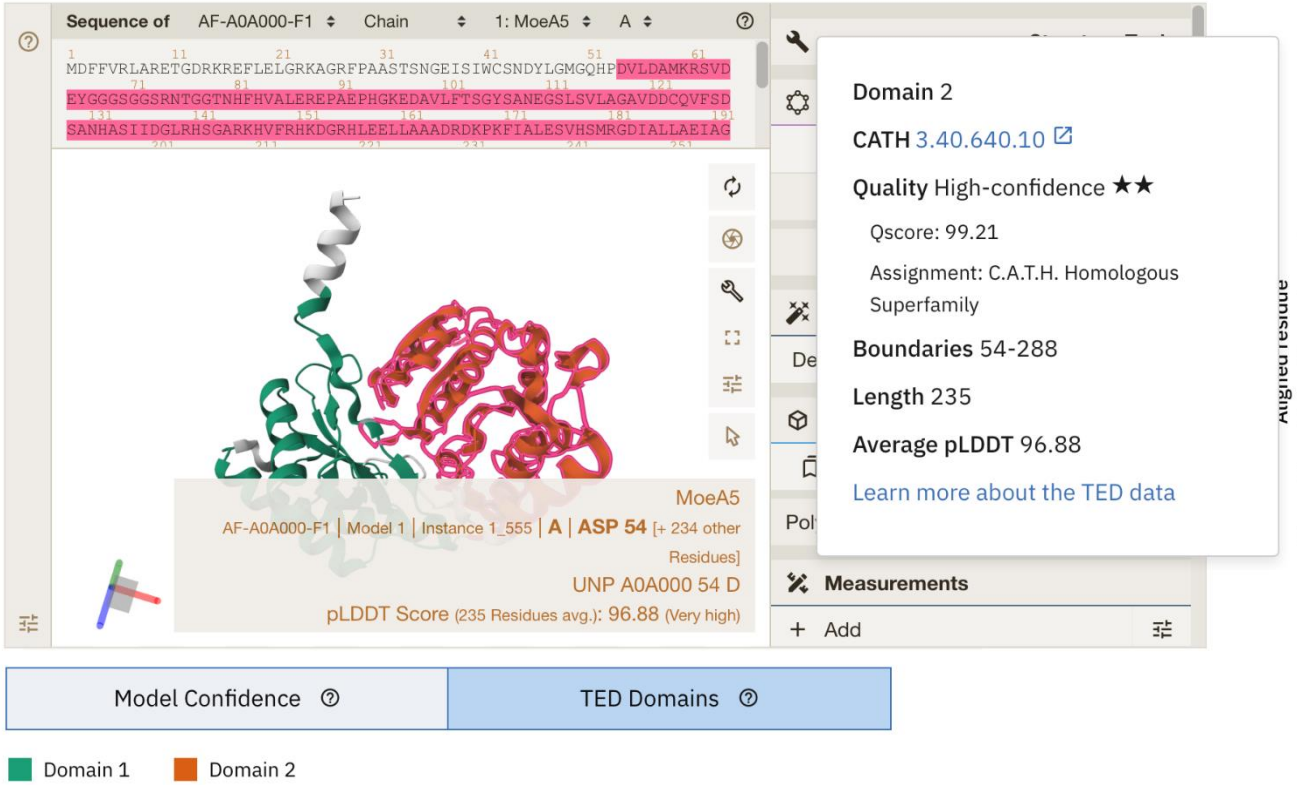


https://alphafold.ebi.ac.uk/

TED domain data in AFDB

Structure viewer

Mol* 3D Viewer



Model Confidence | TED Domains

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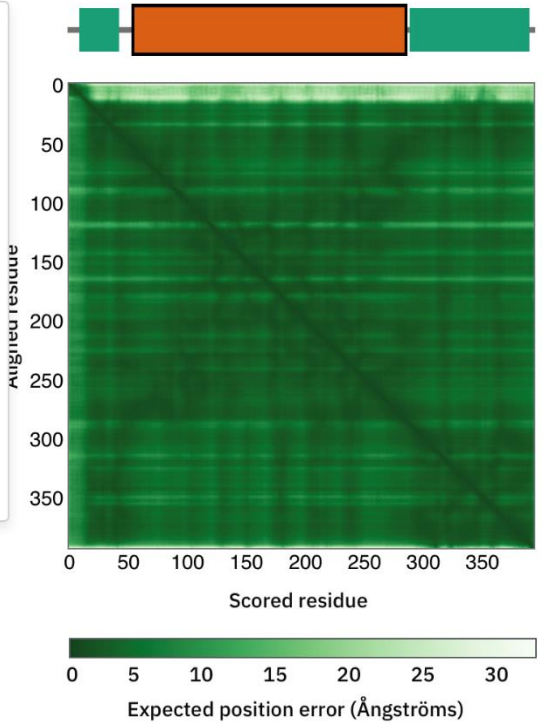

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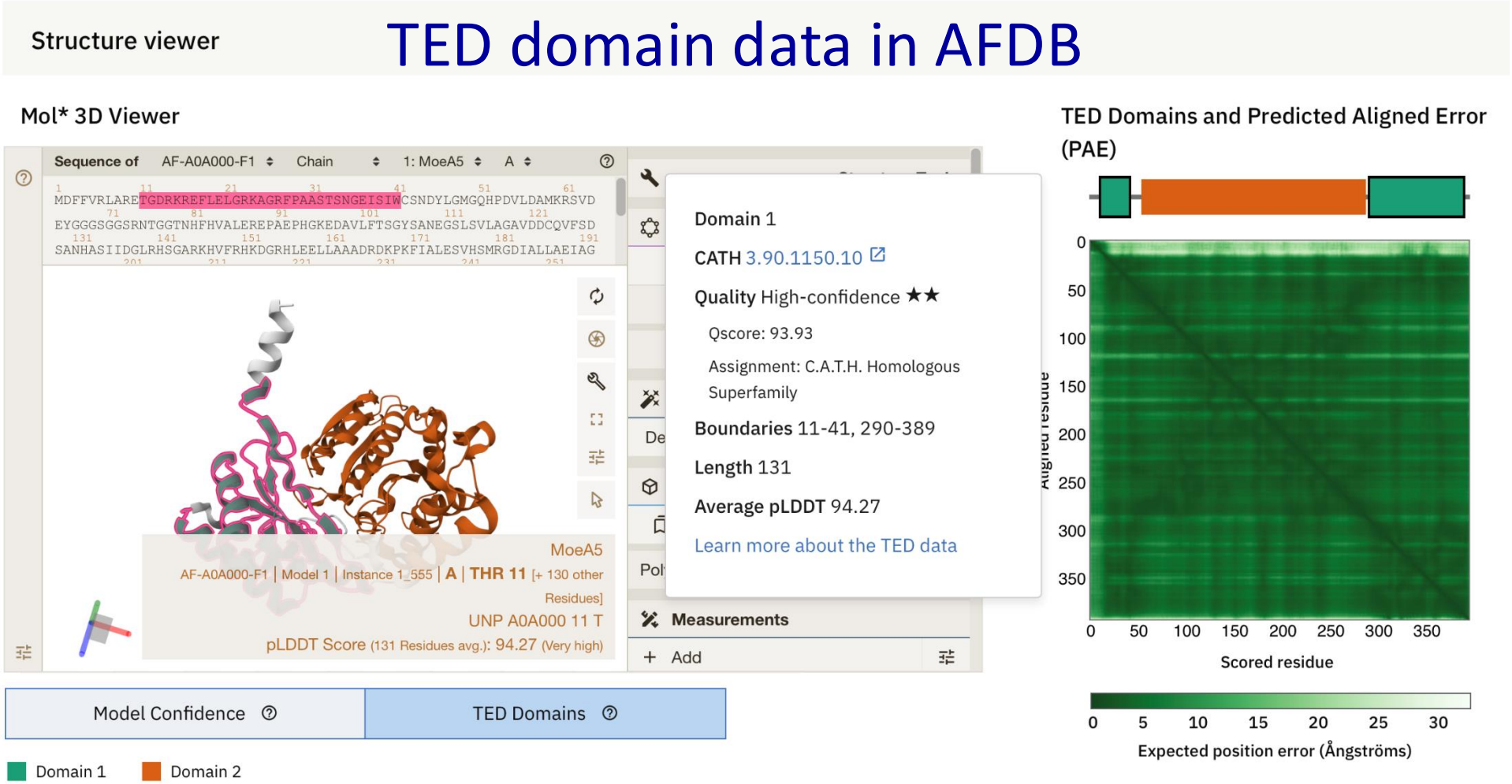


https://alphafold.ebi.ac.uk/

Acknowledgements

Orengo Group

Christine Orengo

Ian Sillitoe (CATH)
Nicola Bordin (CATH)
Vaishali Waman (CATH)

Jude Wells

Jones Group

David Jones

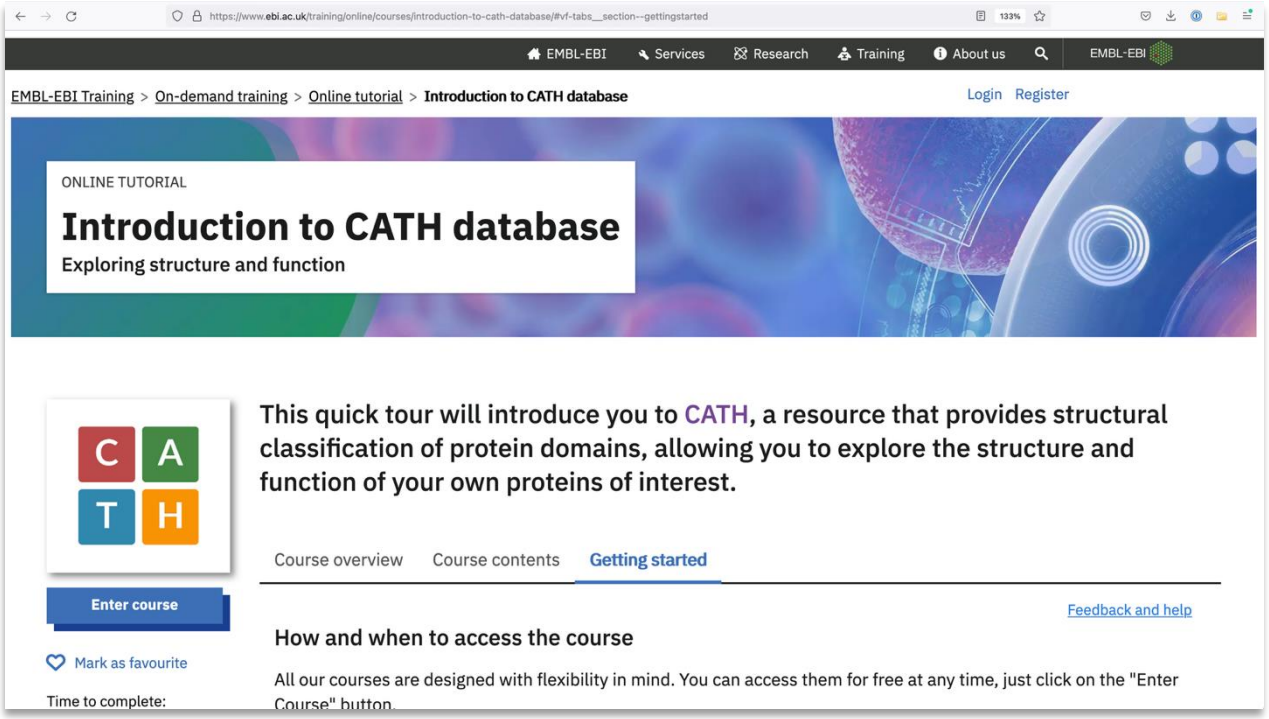
Andy Lau
Shaun Kandathil

Others

Janet Thornton (EBI)
Roman Laskowski (EBI)

BBSRC
Wellcome Trust

Tutorial



[Tutorial - https://www.ebi.ac.uk/training/online/courses/introduction-to-cath-database](https://www.ebi.ac.uk/training/online/courses/introduction-to-cath-database)
(Search: “EBI Training CATH database”)
[CATH - https://www.cathdb.info](https://www.cathdb.info)
[TED -- https://ted.cathdb.info](https://ted.cathdb.info)
[AlphaFoldDB -- https://alphafold.ebi.ac.uk](https://alphafold.ebi.ac.uk)

Questions?

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